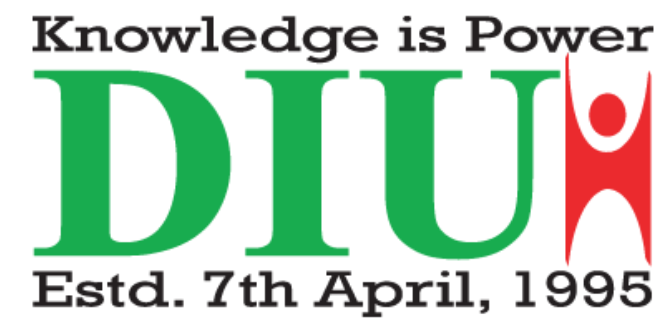




DATA STRUCTURE LAB

COURSE CODE: 0613-104

TALKSPACE

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INTRODUCTION

TalkSpace is a console-based chat system where two users take turns sending messages. The application stores messages in a file, displays the chat history and provides options to search, edit, delete, clear history and exit the chat. The project demonstrates basic programming concepts such as user input, loops, arrays and vectors, file handling and string manipulation



DEPLOYED FEATURES

- **Bi-polar conversation between users.**
- **Unique username for users.**
- **Saving Chat History:** Sent messages are stored & displayed.
- **Clear History:** Users can clear the chat history during the chat session.
- **Exit conversation:** The chat session can be ended by typing 'exit'.
- **Search Messages:** Specific text can be found using the 'search' command.
- **Edit Message:** Last message can be edited by typing 'Edit'.
- **Delete conversation:** Last message can be deleted by typing 'delete'.



FEATURES OVERVIEW

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Custom Username

- ✔ **int main()** saves the custom username using Loop statements for further use.

```
cout << "Enter name for User 1: ";
getline(cin, user1);

while (true) {
    cout << "Enter name for User 2: ";
    getline(cin, user2);
    if (user2 != user1) break;
    cout << "Username already used.\n";
}
```

```
PS C:\Users\User> cd 'c:\Users\User\Desktop\VScode\Code\output'
PS C:\Users\User\Desktop\VScode\Code\output> & .\'TalkSpace.exe'
Enter name for User 1: Ruyaif
Enter name for User 2: Iffat
```

```

      1 1      1
0      1      0 0
1      0 0      1 1      1 0      0      1
0 0 0 1      0 0      0 0 1      0      0      0
1 0 1 0 1 0 1 0 1      1 1 0      1 1      1 1
0 0 0 1 1 1 0 0 0 0 0 0 0 0 0 0 1 0 0
1 1 0 0 1 0 1 1 1 1 1 1 1 0 0 0 1 1
1 0 0 1 0 1 0 0 0 0 0 0 1 0 0 0 1 0 0
1 1 1 0 0 0 0 0 0 0 0 0 1 1 1 0 0 0
1 0 0 1 1 0 0 0 1 1 1 0 1 1 0 0 1 1 0
0 1 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 1 0
0 0 1 0 1 0 1 1 0 1 0 1 0 1 0 0 1 0 1
0 1 0 0 0 1 1 1 1 0 1 0 0 1 0 1 1 1
1 0 0 1 1 1 1 0 0 1 0 1 0 0 1 1 1 0 1
1 0 0 0 1 1 1 0 1 0 1 0 1 0 0 0 1 1 1
1 1 1 0 0 1 0 1 0 1 0 1 0 1 0 1 0 0 0
0      0 0 0 0 0 1 0 1 0 1 0 1 0 1 0 1 0
1      1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1

```

FEATURES OVERVIEW

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Display Chat History

- ✔ **displayChatHistory()** displays the chat history stored in an array.

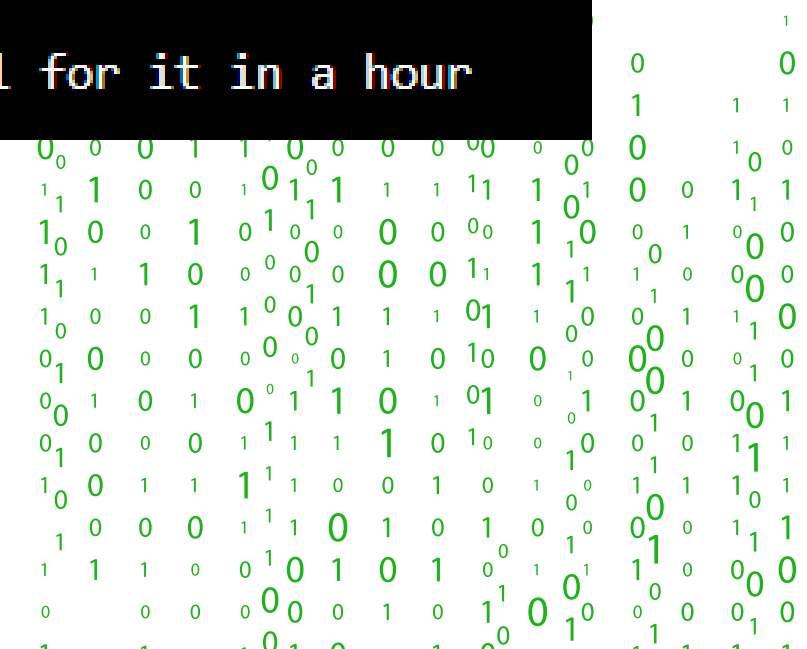
```
void displayChatHistory(const vector<string>& chatHistory) {
    cout << "\n--- Chat History ---" << endl;

    if (chatHistory.empty()) {
        cout << "No messages yet." << endl;
    } else {
        for (const string& msg : chatHistory) {
            cout << msg << endl;
        }
    }
    cout << "-----" << endl;
}
```

```
                                TalkSpace (Persistent Chat)
Commands: exit | clear | edit | delete | search
```

```
--- Chat History ---
No messages yet.
-----
Ruyaif: Hey there? up for tea today?

--- Chat History ---
Ruyaif: Hey there? up for tea today?
-----
Iffat: ya!!! I was about to call for it in a hour
```



FEATURES OVERVIEW

Loading Chat History



loadingChatHistory() loads previous messages from the saved chat history.

```
void loadChatHistory(vector<string>& chatHistory) {
    ifstream file(FILE_NAME);
    string line;

    while (getline(file, line)) {
        chatHistory.push_back(line);
    }
    file.close();
}
```

```

--- Chat History ---
No messages yet.
-----
Ruyaif: Hey there? up for tea today?

--- Chat History ---
Ruyaif: Hey there? up for tea today?
-----
Iffat: ya!!! I was about to call for it in a hour

```

[illegible]

FEATURES OVERVIEW

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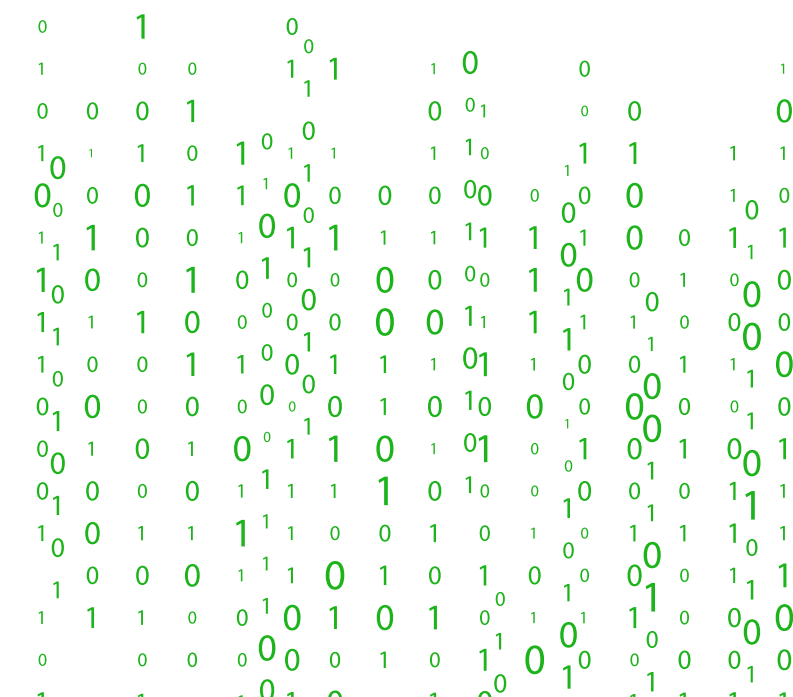
Saving Chat History



savingChatHistory() saves sent messages to the chat history.

```
void saveChatHistory(const vector<string>& chatHistory) {  
    ofstream file(FILE_NAME);  
    for (const string& msg : chatHistory) {  
        file << msg << endl;  
    }  
    file.close();  
}
```

```
--- Chat History ---  
Ruyaif: Hey there? up for tea today?  
Iffat: ya!!! I was about to call for it in a hour  
-----  
Ruyaif: GREAT THEN!!! I will call you at 7:30 pm  
  
--- Chat History ---  
Ruyaif: Hey there? up for tea today?  
Iffat: ya!!! I was about to call for it in a hour  
Ruyaif: GREAT THEN!!! I will call you at 7:30 pm  
-----
```



FEATURES OVERVIEW

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Clearing Chat History

- ✓ **clearChatHistory()** clears the chat history by resetting the message count.

```
void clearChatHistory(vector<string>& chatHistory) {  
    chatHistory.clear();  
    saveChatHistory(chatHistory);  
    cout << "Chat history cleared." << endl;  
}
```

```
--- Chat History ---  
Ruyaif: Hey there? up for tea today?  
Ruyaif: GREAT THEN!!! I will call you at 7:30 pm  
Iffat: Okay!!!!  
-----  
Ruyaif: clear  
Chat history cleared.  
  
--- Chat History ---  
No messages yet.
```



FEATURES OVERVIEW

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Searching text

- ✔ **SearchMessages()** searches specific text from the conversation using getline function, loops & conditional statement.

```
void searchMessages(const vector<string>& chatHistory) {
    string keyword;
    cout << "Enter keyword to search: ";
    getline(cin, keyword);

    bool found = false;
    cout << "\n--- Search Results ---" << endl;

    for (const string& msg : chatHistory) {
        if (msg.find(keyword) != string::npos) {
            cout << msg << endl;
            found = true;
        }
    }

    if (!found) {
        cout << "No matching messages found." << endl;
    }
    cout << "-----" << endl;
}
```

```
--- Chat History ---
Rafid: We have a match on this Friday.
Sadat: When?
Rafid: 16:00. Make sure to arrive before 15:30!
Sadat: ALRIGHT!!!
```

```
-----
Rafid: search
Enter keyword to search: When?
```

```
--- Search Results ---
Sadat: When?
```

```
-----
--- Chat History ---
Rafid: We have a match on this Friday.
Sadat: When?
Rafid: 16:00. Make sure to arrive before 15:30!
Sadat: ALRIGHT!!!
```

FEATURES OVERVIEW

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Edit the last message

- ✓ **editLastMessage()** edit the last message using string, constant, getline function & loop statements.

```
void editLastMessage(vector<string>& chatHistory, const string& currentUser) {
    for (int i = chatHistory.size() - 1; i >= 0; i--) {
        if (chatHistory[i].find(currentUser + ": ") == 0) {
            string newMsg;
            cout << "Enter new message: ";
            getline(cin, newMsg);
            chatHistory[i] = currentUser + ": " + newMsg;
            saveChatHistory(chatHistory);
            cout << "Message edited." << endl;
            return;
        }
    }
    cout << "No message found to edit." << endl;
}
```

```
--- Chat History ---
Batch Cordinator: 5th Semester Final examination is set to start from 12nd February,2026
Batch Representative: NOTED.
-----
Batch Cordinator: edit
Enter new message: The starting date of 5th Semester final examination will be announce later considering the national parliament election date.
Message edited.
```

FEATURES OVERVIEW

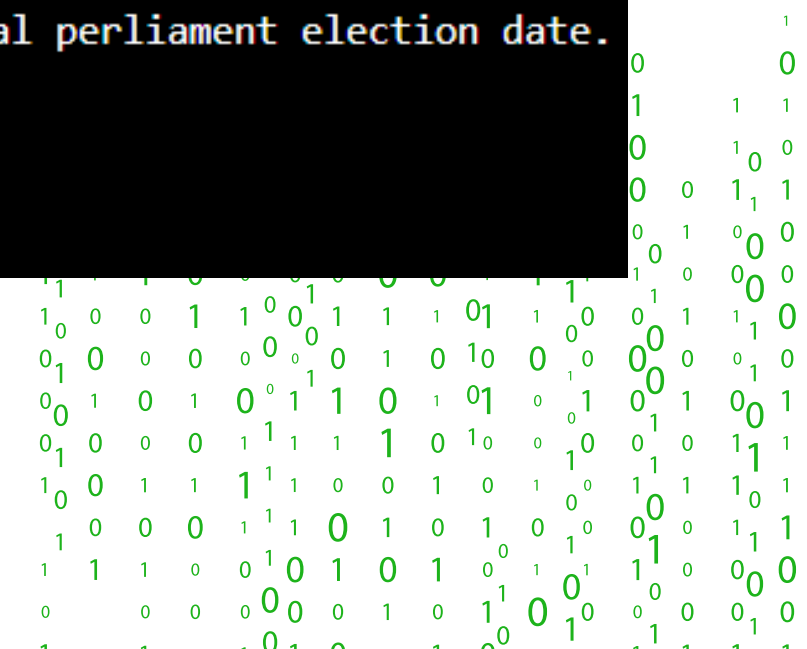
10

Delete the last message

✔ **deleteLastMessage()** edit the last message using string, constant, getline function & loop statements.

```
void deleteLastMessage(vector<string>& chatHistory, const string& currentUser) {
    for (int i = chatHistory.size() - 1; i >= 0; i--) {
        if (chatHistory[i].find(currentUser + ": ") == 0) {
            chatHistory.erase(chatHistory.begin() + i);
            saveChatHistory(chatHistory);
            cout << "Message deleted." << endl;
            return;
        }
    }
    cout << "No message found to delete." << endl;
}
```

```
--- Chat History ---
Batch Cordinator: The starting date of 5th Semester final examination will be announce later considering the national perliament election date.
Batch Representative: NOTED.
-----
Batch Cordinator: delete
Message deleted.
```



CONCLUSION

The project implements a minimal chatting system for 2 users, with features like: saving conversations, searching message, editing & deleting message, clearing chat history & logging out sessions.

```

      1 1      1
0      1      0 0
1      0 0      1 1      1 0      0
0 0 0 1      0 0      0 0 1
1 0 1 0 1 0 1 0 1 1 1 0 1
0 0 0 1 1 0 0 0 0 0 0 0 0 0
1 1 0 0 1 0 1 1 1 1 1 1 1 0
1 0 0 1 0 1 0 0 0 0 0 1 1 0
1 1 1 0 0 0 0 0 0 0 0 1 1 1
1 0 0 1 1 0 0 1 1 1 1 0 1 0
0 1 0 0 0 0 0 0 0 1 0 1 0 0
0 0 1 0 1 0 1 1 0 1 0 1 0 1
0 1 0 0 0 1 1 1 1 0 1 0 0 1
1 0 0 1 1 1 1 0 0 1 0 1 0 0
1 0 0 0 1 1 1 0 1 0 1 0 1 0
1 1 1 0 0 1 0 1 0 1 0 1 0 1
0 0 0 0 0 0 0 1 0 1 0 1 0
1 1 1 1 0 1 0 1 0 1 0 1 0
1 0 0 0 1 0 1 0 0 1 0 1 0

```

ANY QUESTIONS ?

